# Vision-Based BEV Perception for Energy-Optimal Autonomous Driving using PPO

Internal Technical Specification

March 11, 2026

## Contents


**1 Overview and Philosophy** **3**
- 1.1 Core Philosophy . . . 3
- 1.2 What is and is not privileged information . . . 3

**2 Camera Setup** **3**
- 2.1 Camera Configuration . . . 3
- 2.2 Intrinsic Parameters . . . 4
- 2.3 Stereo Pair . . . 4

**3 Stereo Depth Estimation** **4**
- 3.1 Depth Precision Analysis . . . 4
- 3.2 Disparity to Depth . . . 5
- 3.3 SGM Stereo Matching Parameters . . . 5

**4 Object Detection** **5**

**5 BEV Grid Construction** **6**
- 5.1 Grid Specification . . . 6
- 5.2 Grid Values . . . 6
- 5.3 Grid Visualisation . . . 7
- 5.4 Coordinate Mapping: World to Grid . . . 7
- 5.5 Camera Frame to Ego Frame Rotation . . . 7
- 5.6 Monocular Depth (Side and Rear Cameras) . . . 8
- 5.7 Physical Size Estimation . . . 8
- 5.8 Ego Car and Route Drawing . . . 8

**6 State and Action Space** **8**
- 6.1 Observation . . . 8
  - 6.1.1 BEV Grid . . . 8
  - 6.1.2 Scalar State . . . 9
- 6.2 Action Space . . . 9

**7 Physics-Grounded Energy Model** **9**
- 7.1 Longitudinal Dynamics . . . 9
- 7.2 Force Components . . . 10
- 7.3 Power and Energy . . . 10
- 7.4 Vehicle Parameters . . . 10

**8 Reward Function** **10**
8.1 Per-Step Reward . . . 10
8.2 Terminal Rewards . . . 11
8.3 Reward Weights . . . 11

**9 Policy Architecture** **11**
9.1 Overview . . . 11
9.2 CNN Branch . . . 12
9.3 MLP Branch . . . 12
9.4 Fusion and Action Head . . . 12

**10 Training Pipeline** **12**
10.1 Overview . . . 12
10.2 Phase 1 — Behavioural Cloning . . . 12
10.3 Phase 2 — RL Finetuning, No Traffic . . . 12
10.4 Phase 3 — RL Finetuning, Traffic + Domain Randomisation . . . 13
10.5 PPO Hyperparameters . . . 13

**11 CARLA Environment Details** **13**
11.1 Simulation Settings . . . 13
11.2 Reset Sequence . . . 13
11.3 Termination Conditions . . . 14

**12 Key Design Decisions Summary** **14**

# 1 Overview and Philosophy

This document describes the full architecture, perception pipeline, physics model, and training strategy for our autonomous driving agent trained in CARLA using Proximal Policy Optimisation (PPO).

## 1.1 Core Philosophy

The central design principle is:

> *"If a reinforcement learning agent can learn to play Atari games from raw pixels alone with no prior knowledge, it can learn to navigate an urban environment from a structured Bird's Eye View (BEV) representation — provided a well-designed reward function."*

Atari DQN had to learn what objects are, where they are, and how they move purely from RGB pixels. Our agent receives a semantically structured 64×64 occupancy grid where obstacles, route, and road boundaries are already spatially grounded. This is a strictly easier visual task.

## 1.2 What is and is not privileged information

A key design constraint is that no privileged simulator information is used for NPC (other vehicle) perception. All obstacle detection is vision-only via cameras and YOLO.

| **Signal** | **Source** | **Realistic?** |
|---|---|---|
| Frenet state ($s$, $d$, $\kappa$, $\theta_{\text{err}}$, $d_{\text{goal}}$) | HD map + GPS | ✓ Yes |
| Route centreline | HD map waypoints | ✓ Yes |
| Road boundaries | HD map lane edges | ✓ Yes |
| NPC positions and sizes | YOLO + stereo geometry | ✓ Yes |
| NPC exact world positions | CARLA simulator | × No — not used |

Table 1: Information sources and their real-world validity.

HD map usage for ego localisation is consistent with all production AV stacks (Waymo, Cruise, etc.) which localise against pre-mapped environments using GPS and LiDAR.

# 2 Camera Setup

## 2.1 Camera Configuration

Four cameras are mounted on the ego vehicle facing front, left, right, and rear. All cameras share identical intrinsic parameters.

| **Camera** | **x (m)** | **z (m)** | **Yaw** | **Pitch** | **Role** |
|---|---|---|---|---|---|
| Front | +2.0 | 1.4 | 0 | −5 | Stereo left + primary |
| Left | 0.0 | 1.4 | −90 | −5 | Monocular side |
| Right | 0.0 | 1.4 | +90 | −5 | Monocular side |
| Rear | −2.0 | 1.4 | 180 | −5 | Monocular rear |

Table 2: Camera mounting positions and orientations on ego vehicle. $x$ = forward offset, $z$ = height above ground.

### 2.2 Intrinsic Parameters

All cameras share the following intrinsic parameters, derived from the stereo depth precision requirement of $\sigma_Z \leq 0.36\,\text{m}$ at $Z = 30\,\text{m}$:

| Parameter | Value |
|---|---|
| Image width $w$ | 1200 px |
| Image height $h$ | 900 px |
| Field of view (horizontal) $\text{FOV}_h$ | 60 |
| Focal length $f$ | 1000 px |
| Height above ground $H$ | 1.4 m |
| Downward pitch $\theta_{\text{pitch}}$ | $-5$ |

Table 3: Camera intrinsic and mounting parameters.

The focal length is derived from resolution and FOV:

$$f = \frac{w/2}{\tan(\text{FOV}_h/2)} = \frac{600}{\tan(30)} = \frac{600}{0.577} \approx 1000\,\text{px} \tag{1}$$

### 2.3 Stereo Pair

Depth estimation uses a stereo pair consisting of two front-facing cameras separated by a horizontal baseline:

$$B = 0.5\,\text{m} \tag{2}$$

The left and right cameras of the stereo pair are mounted symmetrically 0.25 m either side of the vehicle centreline at the front. The remaining cameras (left-side, right-side, rear) use monocular ground-plane projection for depth since high-precision depth to the sides and rear is less critical for control decisions.

## 3 Stereo Depth Estimation

### 3.1 Depth Precision Analysis

Stereo depth error grows quadratically with distance:

$$\sigma_Z = \frac{Z^2 \cdot \sigma_d}{f \cdot B} \tag{3}$$

where:

- $\sigma_Z$ = depth error (metres)
- $Z$ = target depth (metres)
- $\sigma_d$ = disparity error (pixels) — approximately 0.2 px with Semi-Global Matching (SGM)
- $f$ = focal length = 1000 px
- $B$ = baseline = 0.5 m

With our parameters:

$$\sigma_Z = \frac{Z^2 \times 0.2}{1000 \times 0.5} = \frac{Z^2}{2500} \tag{4}$$

| Distance | Depth Error | Relative Error |
|---|---|---|
| 10 m | 0.04 m | 0.4% |
| 20 m | 0.16 m | 0.8% |
| 30 m | 0.36 m | 1.2% |

Table 4: Stereo depth precision at key distances. Sub-metre accuracy throughout the operating range.

### 3.2 Disparity to Depth

For each pixel with valid disparity $d_{\text{px}}$:

$$Z = \frac{f \cdot B}{d_{\text{px}}} = \frac{500}{d_{\text{px}}} \tag{5}$$

Lateral and vertical 3D position:

$$X = \frac{(u - c_x) \cdot Z}{f}, \qquad Y = \frac{(v - c_y) \cdot Z}{f} \tag{6}$$

where $(c_x, c_y)$ is the image principal point (image centre).

### 3.3 SGM Stereo Matching Parameters

| Parameter | Value |
|---|---|
| Algorithm | Semi-Global Matching (SGM) |
| `numDisparities` | 192 (divisible by 16 — OpenCV requirement) |
| `blockSize` | 5 |
| $P_1$ | $8 \times 3 \times 5^2 = 600$ |
| $P_2$ | $32 \times 3 \times 5^2 = 2400$ |
| `disp12MaxDiff` | 1 |
| `uniquenessRatio` | 10 |
| Mode | `STEREO_SGBM_MODE_SGBM_3WAY` |
| Depth range | 1 m – 30 m |

Table 5: OpenCV StereoSGBM configuration.

The choice of `numDisparities` = 192 covers depths from approximately 3 m to 30 m:

$$d_{\text{px}}^{\max} = \frac{f \cdot B}{Z_{\min}} = \frac{500}{3} \approx 167\,\text{px} \quad \Rightarrow \quad \text{round up to 192 (multiple of 16)} \tag{7}$$

## 4 Object Detection

YOLO v8n runs independently on each of the four camera images at every timestep. Detections are filtered to obstacle classes only (cars, trucks, buses, motorcycles, pedestrians — COCO class IDs 2, 3, 5, 7, 0) with a confidence threshold of 0.4.

Each detection produces:

$$\{[x_1, y_1, x_2, y_2],\ \text{confidence},\ \text{class_id}\} \tag{8}$$

The bounding box bottom-centre pixel is used for ground-plane projection and the full box dimensions are used for physical size estimation.

# 5 BEV Grid Construction

## 5.1 Grid Specification

| Property | Value |
|---|---|
| Grid size | $64 \times 64$ pixels |
| Lateral range | $\pm 10$ m (columns 0–63) |
| Forward range | $+17.5$ m ahead (rows 0–55) |
| Rear range | $-2.5$ m behind (rows 57–63) |
| Ego car position | row = 56, col = 32 |
| Metres per pixel | $20\,\text{m}/64 = 0.3125\,\text{m/px}$ |

Table 6: BEV grid specification. Ego is offset toward the bottom to allocate more grid space to the forward direction.

## 5.2 Grid Values

| Value | Meaning |
|---|---|
| 0.0 | Free drivable space — ego car can go here |
| 0.5 | Route centreline — from HD map waypoints |
| 0.8 | Road boundary — from HD map lane edges |
| 1.0 | Physically occupied — ego footprint or any obstacle |

Table 7: BEV grid cell values and their semantic meaning.

The key property of this representation: **free space (0.0) means the ego car physically fits there**. Obstacles are drawn at their true physical size, not as point estimates. This eliminates the need for inflation heuristics used in classical costmaps.

### 5.3 Grid Visualisation

Listing 1: ASCII representation of the 64×64 BEV grid. Each character represents a region of the grid.

```
Col:  0               32              63
      |<---- 10m ---->|<---- 10m ---->|

Row 0  . . . . . . . . . . . . . . . .   <- far ahead (+17.5m)
       . . . . . . . . . . . . . . . .
       . . . | . . . . . . . | . . .   <- road boundary (0.8)
       . . . | . - - - . . . | . . .   <- route centreline (0.5)
       . . . | . - - - . . . | . . .
       . . . | . - - - . . . | . . .
       . XXX | . - - - . . . | . . .   <- left: obstacle (1.0)
       . XXX | . - - - . . . | . . .   <- true physical rectangle
       . . . | . -XXX- . . . | . . .   <- front: NPC on route (1.0)
       . . . | . -XXX- . . . | . . .
       . . . | . - - - . . . | . . .
       . . . | . - - - . XXX | . . .   <- right: car to the right
       . . . | . - - - . XXX | . . .
       . . . | . - - - . . . | . . .
       . . . | . - - - . . . | . . .
Row 56 . . . | . =[E]= . . . | . . .   <- ego car footprint at row=56
       . . . | . . . . . . . | . . .   <- behind ego
       . . . | . . . . . . . | . . .
       . XXXXXXX . . . . . . . . . .   <- rear: tailgating car (1.0)
Row 63 . . . . . . . . . . . . . . . .   <- near behind (-2.5m)

Legend:
  [E] = ego car true footprint (1.0), 2m x 4.5m
  XXX = obstacle true footprint (1.0)
  ---  = route centreline (0.5)
  |    = road boundary (0.8)
  .    = free drivable space (0.0)
```

### 5.4 Coordinate Mapping: World to Grid

Given a point at ego-frame coordinates $(X_{\text{ego}}, Z_{\text{ego}})$ where $X_{\text{ego}}$ is lateral (right = positive) and $Z_{\text{ego}}$ is forward (ahead = positive):

$$\text{col} = \left\lfloor \frac{X_{\text{ego}} + 10.0}{20.0} \times 64 \right\rfloor \tag{9}$$

$$\text{row} = \left\lfloor \left(1 - \frac{Z_{\text{ego}} + 2.5}{20.0}\right) \times 64 \right\rfloor \tag{10}$$

### 5.5 Camera Frame to Ego Frame Rotation

Each camera has a known yaw offset from the vehicle's forward direction. Detections in camera-local $(X_{\text{cam}}, Z_{\text{cam}})$ are rotated into ego frame via:

| **Camera** | $X_{\mathbf{ego}}$ | $Z_{\mathbf{ego}}$ |
|---|---|---|
| Front (0) | $X_{\text{cam}}$ | $Z_{\text{cam}}$ |
| Left ($-90$) | $-Z_{\text{cam}}$ | $X_{\text{cam}}$ |
| Right ($+90$) | $Z_{\text{cam}}$ | $-X_{\text{cam}}$ |
| Rear (180) | $-X_{\text{cam}}$ | $-Z_{\text{cam}}$ |

Table 8: Camera-local to ego-frame coordinate rotation for each camera.

### 5.6 Monocular Depth (Side and Rear Cameras)

For non-stereo cameras, depth is estimated via ground-plane projection. Given camera height $H$, pitch $\theta_{\text{pitch}}$, and the vertical pixel offset $\Delta v$ of the bounding box bottom from image centre:

$$\varphi = \theta_{\text{pitch}} + \Delta v \cdot \frac{\text{FOV}_v}{h_{\text{img}}} \tag{11}$$

$$Z = \frac{H}{\tan(\varphi)} \tag{12}$$

Lateral offset:

$$\beta = \Delta u \cdot \frac{\text{FOV}_h}{w_{\text{img}}}, \qquad X = Z \cdot \tan(\beta) \tag{13}$$

where $\Delta u$ is the horizontal pixel offset of the bounding box centre from the image centre.

### 5.7 Physical Size Estimation

Object physical dimensions are estimated from bounding box size and depth:

$$w_{\text{physical}} = \frac{w_{\text{bbox}} \cdot Z}{f}, \qquad l_{\text{physical}} = \frac{h_{\text{bbox}} \cdot Z}{f} \tag{14}$$

where $w_{\text{bbox}}$ and $h_{\text{bbox}}$ are bounding box width and height in pixels, and $f = 1000\,\text{px}$ is the focal length.

The object is then drawn on the BEV grid as a rectangle of size $(w_{\text{physical}} \times l_{\text{physical}})$ centred at the projected $(X_{\text{ego}}, Z_{\text{ego}})$ position, converted to pixel dimensions using the grid's metres-per-pixel ratio.

### 5.8 Ego Car and Route Drawing

- **Ego footprint:** 2.0 m wide $\times$ 4.5 m long rectangle centred at (row = 56, col = 32), value = 1.0
- **Route centreline:** HD map waypoints transformed to ego frame, drawn as value = 0.5
- **Road boundaries:** HD map lane edges transformed to ego frame, drawn as value = 0.8

## 6 State and Action Space

### 6.1 Observation

The full observation has two components:

#### 6.1.1 BEV Grid

A $64 \times 64$ float32 array as described above. Flattened to 4096-dim for the MLP branch or kept as 2D for the CNN branch.

#### 6.1.2 Scalar State

$$\mathbf{s}_t = [v_t,\ s_t,\ d_t,\ \theta_{\text{err},t},\ \kappa_{\text{ahead},t},\ d_{\text{goal},t}] \tag{15}$$

| Symbol | Meaning | Source | Normalisation |
|---|---|---|---|
| $v_t$ | Ego speed (m/s) | IMU | $\div 30.0$ |
| $s_t$ | Route progress (m) | HD map + GPS | $\div 1000.0$ |
| $d_t$ | Lateral deviation (m) | HD map + GPS | $\div 5.0$ |
| $\theta_{\text{err},t}$ | Heading error (rad) | HD map + IMU | $\div \pi$ |
| $\kappa_{\text{ahead},t}$ | Upcoming curvature | HD map | $\times 10$ |
| $d_{\text{goal},t}$ | Distance to goal (m) | HD map | $\div 1000.0$ |

Table 9: Scalar state components, sources, and normalisation.

$\kappa$ (kappa) is road curvature $= 1/r$ where $r$ is the curve radius in metres. On a straight road $\kappa \approx 0$; on a sharp corner $\kappa$ is large. It gives the policy preview of upcoming geometry.

### 6.2 Action Space

$$\mathbf{a}_t = [\delta_t,\ u_t,\ b_t] \tag{16}$$

| Symbol | Meaning | Range |
|---|---|---|
| $\delta_t$ | Steering | $[-0.3,\ +0.3]$ |
| $u_t$ | Throttle | $[0.0,\ 1.0]$ |
| $b_t$ | Brake | $[0.0,\ 1.0]$ |

Table 10: Continuous action space.

## 7 Physics-Grounded Energy Model

### 7.1 Longitudinal Dynamics

Vehicle longitudinal dynamics including wheel rotational inertia under the no-slip condition ($a_{\text{wheel}} = a/r$):

$$\left(m + m_{\text{payload}} + \frac{4I}{r^2}\right) \cdot a_t = F_{\text{engine}} - F_{\text{drag}} - F_{\text{friction}} - (m + m_{\text{payload}}) \cdot g \cdot \sin(\theta_{\text{road}}) \tag{17}$$

where $I$ is the moment of inertia of one wheel and $r$ is the wheel radius. The no-slip condition gives angular acceleration $\alpha = a_t/r$, so the 4-wheel rotational inertia contributes an effective additional mass of $4I/r^2$ to the system.

Rearranging for engine force (what the engine must provide):

$$F_{\text{engine}} = F_{\text{inertia}} + F_{\text{drag}} + F_{\text{friction}} + F_{\text{gravity}} \tag{18}$$

### 7.2 Force Components

$$F_{\text{inertia}} = \left(m + \frac{4I}{r^2}\right) \cdot a_t \tag{19}$$

$$F_{\text{drag}} = \tfrac{1}{2}\rho C_d A v_t^2 \tag{20}$$

$$F_{\text{friction}} = C_{rr} \cdot m \cdot g \cdot \cos(\theta_{\text{road}}) \tag{21}$$

$$F_{\text{gravity}} = m \cdot g \cdot \sin(\theta_{\text{road}}) \tag{22}$$

where $\theta_{\text{road}}$ is the road gradient angle in radians, extracted from CARLA waypoint pitch. The wheel inertia correction evaluates to:

$$\frac{4I}{r^2} = \frac{4 \times 1.5}{0.35^2} \approx 48.98\,\text{kg} \tag{23}$$

giving an effective total inertial mass of $1800 + 48.98 \approx 1849\,\text{kg}$ — a correction of $\approx 2.7\%$.

**Code simplification:** In the implementation, the $4I/r^2$ term is omitted and $F_{\text{inertia}} = m \cdot a_t$ is used directly. This 2.7% correction is smaller than the physics noise introduced by CARLA's gear-change velocity discontinuities. This simplification is noted as a known limitation.

### 7.3 Power and Energy

$$P_{\text{engine}} = F_{\text{engine}} \cdot v_t \tag{24}$$

$$E_t = \max(0,\ P_{\text{engine}} \cdot \Delta t) \tag{25}$$

The $\max(0, \cdot)$ handles downhill coasting where $F_{\text{engine}}$ is negative (engine does no work, gravity does the work).

### 7.4 Vehicle Parameters

| Parameter | Symbol | Value |
|---|---|---|
| Vehicle mass | $m$ | $1800\,\text{kg}$ |
| Payload | $m_{\text{payload}}$ | $0\,\text{kg}$ |
| Drag coefficient | $C_d$ | 0.23 |
| Frontal area | $A$ | $2.22\,\text{m}^2$ |
| Rolling resistance | $C_{rr}$ | 0.01 |
| Air density | $\rho$ | $1.225\,\text{kg/m}^3$ |
| Gravity | $g$ | $9.81\,\text{m/s}^2$ |
| Wheel moment of inertia | $I$ | $1.5\,\text{kg m}^2$ |
| Wheel radius | $r$ | $0.35\,\text{m}$ |
| Effective inertial mass | $m + 4I/r^2$ | $\approx 1849\,\text{kg}$ |
| Timestep | $\Delta t$ | $0.04\,\text{s}$ (25 Hz) |

Table 11: Physics model parameters (Tesla Model 3 approximations).

## 8 Reward Function

### 8.1 Per-Step Reward

$$R_t = r_{\text{progress}} + r_{\text{speed}} + r_{\text{energy}} + r_{\text{lateral}} + r_{\text{heading}} + r_{\text{brake}} + r_{\text{jerk}} + r_{\text{still}} + r_{\text{speed_cap}} + r_{\text{lane}} \tag{26}$$

$$r_{\text{progress}} = \alpha \cdot \Delta s_t \tag{27}$$
$$r_{\text{speed}} = \beta \cdot v_t \cdot \text{lane_ok} \tag{28}$$
$$r_{\text{energy}} = -\lambda_E \cdot E_t \tag{29}$$
$$r_{\text{lateral}} = -\lambda_d \cdot d_t^2 \tag{30}$$
$$r_{\text{heading}} = -\lambda_\theta \cdot \theta_{\text{err},t}^2 \tag{31}$$
$$r_{\text{brake}} = -\lambda_b \cdot b_t \tag{32}$$
$$r_{\text{jerk}} = -\lambda_j \cdot (\Delta a_t)^2 \tag{33}$$
$$r_{\text{still}} = r_{\text{still_penalty}} \text{ if } v_t < 0.5\,\text{m/s} \text{ and } \Delta s < 0.01\,\text{m} \tag{34}$$
$$r_{\text{speed_cap}} = -\lambda_{\text{sc}} \cdot \max(0,\, v_t - v_{\text{target}})^2 \tag{35}$$
$$r_{\text{lane}} = -\lambda_{\text{lane}} \cdot \max(0,\, |d_t| - d_{\max}) \tag{36}$$

where $\Delta a_t = a_t - a_{t-1}$ is the jerk (change in acceleration), and $\text{lane_ok} = \max(0, 1 - |d_t|/d_{\max})$.

### 8.2 Terminal Rewards

| Event | Terminal Reward |
|---|---|
| Goal reached | $+500$ |
| Collision | $-200$ |
| Lane violation | $-30 \times \lvert d_t \rvert$ |
| Timeout | 0 |

Table 12: Terminal reward values.

### 8.3 Reward Weights

| Weight | Symbol | Value |
|---|---|---|
| Progress | $\alpha$ | 10.0 |
| Speed | $\beta$ | 1.0 |
| Energy | $\lambda_E$ | 0.01 |
| Lateral | $\lambda_d$ | 3.0 |
| Heading | $\lambda_\theta$ | 8.0 |
| Brake | $\lambda_b$ | 0.5 |
| Jerk | $\lambda_j$ | 0.1 |
| Speed cap | $\lambda_{\text{sc}}$ | 0.5 |
| Lane violation | $\lambda_{\text{lane}}$ | 30.0 |
| Still penalty | $r_{\text{still}}$ | $-5.0$ |
| Target speed | $v_{\text{target}}$ | 12.0 m/s |
| Max lateral dev | $d_{\max}$ | 3.0 m |

Table 13: Reward function weights.

## 9 Policy Architecture

### 9.1 Overview

The policy has two input branches that are fused before the action head:

1. **CNN branch** — processes the $64 \times 64$ BEV grid
2. **MLP branch** — processes the 6-dim scalar state

### 9.2 CNN Branch

$$\underbrace{64 \times 64}_{\text{BEV grid}} \xrightarrow{\text{Conv(1,32,3)}} \text{ReLU} \xrightarrow{\text{Conv(32,64,3)}} \text{ReLU} \xrightarrow{\text{MaxPool(2)}} 32{\times}32 \xrightarrow{\text{Conv(64,64,3)}} \text{ReLU} \xrightarrow{\text{MaxPool(2)}} 16{\times}16 \xrightarrow{\text{Flatten}} 1 \tag{37}$$

### 9.3 MLP Branch

$$\underbrace{\mathbb{R}^6}_{\text{scalar}} \xrightarrow{\text{FC(6,64)}} \text{ReLU} \xrightarrow{\text{FC(64,32)}} \text{ReLU} \to \mathbf{h}_{\text{motion}} \in \mathbb{R}^{32} \tag{38}$$

### 9.4 Fusion and Action Head

$$\mathbf{h}_{\text{fused}} = \text{concat}(\mathbf{h}_{\text{spatial}}, \mathbf{h}_{\text{motion}}) \in \mathbb{R}^{160} \tag{39}$$

$$\mathbf{h}_{\text{fused}} \xrightarrow{\text{FC(160,128)}} \text{ReLU} \xrightarrow{\text{FC(128,64)}} \text{ReLU} \xrightarrow{\text{FC(64,3)}} \tanh \to [\delta_t, u_t, b_t] \tag{40}$$

## 10 Training Pipeline

### 10.1 Overview

Training proceeds in three phases following a *curriculum-guided imitation bootstrapping* strategy:

$$\underbrace{\text{BC (no traffic)}}_{\text{Phase 1}} \longrightarrow \underbrace{\text{RL (no traffic + energy)}}_{\text{Phase 2}} \longrightarrow \underbrace{\text{RL (traffic + domain randomisation)}}_{\text{Phase 3}} \tag{41}$$

### 10.2 Phase 1 — Behavioural Cloning

- **Demos:** PS4 controller teleoperation or rule-based expert
- **Objective:** $\mathcal{L}_{\text{BC}} = \text{MSE}(\pi_\theta(\mathbf{s}), \mathbf{a}_{\text{expert}})$
- **Traffic:** none
- **Demo pairs:** 50k–100k
- **Purpose:** seed policy with basic lane-following behaviour so RL does not waste episodes learning to move forward

### 10.3 Phase 2 — RL Finetuning, No Traffic

- **Initialisation:** resume from Phase 1 BC policy
- **Energy penalty:** $\lambda_E = 0.01$ active
- **Traffic:** $n_{\text{vehicles}} = 0$
- **Steps:** 500k–1M
- **Purpose:** learn energy-efficient driving without distraction of traffic

### 10.4 Phase 3 — RL Finetuning, Traffic + Domain Randomisation

- **Initialisation:** resume from Phase 2
- **Traffic:** $n_{\text{vehicles}} = 5$–$10$
- **Domain randomisation:** weather, lighting, NPC density randomised each episode to bridge sim-to-real gap
- **Steps:** 500k–1M
- **Purpose:** learn obstacle avoidance; robust policy for real-world transfer

### 10.5 PPO Hyperparameters

| Parameter | Value |
|---|---|
| Learning rate | $3 \times 10^{-4}$ |
| $n_{\text{steps}}$ | 512 |
| Batch size | 64 |
| Epochs per update | 10 |
| $\gamma$ | 0.99 |
| GAE $\lambda$ | 0.95 |
| Clip range | 0.2 |
| Entropy coeff. | 0.01 |

Table 14: PPO hyperparameters.

## 11 CARLA Environment Details

### 11.1 Simulation Settings

| Setting | Value |
|---|---|
| Town | Town10HD_Opt |
| Timestep $\Delta t$ | 0.04 s (25 Hz, synchronous mode) |
| Vehicle | `vehicle.tesla.model3` |
| Max episode steps | 2000 |

### 11.2 Reset Sequence

CARLA's automatic transmission takes ≈10 ticks to engage gear 1 from cold. The reset sequence ensures the drivetrain is engaged before the episode begins:

Listing 2: Reset drivetrain warmup sequence.

```
# Step 1 - warm up drivetrain (20 ticks to reliably engage gear 1)
for _ in range(20):
    vehicle.apply_control(VehicleControl(throttle=0.5, brake=0.0,
                                         hand_brake=False))
    world.tick()

# Step 2 - brake to standstill
for _ in range(10):
    vehicle.apply_control(VehicleControl(throttle=0.0, brake=1.0,
                                         hand_brake=False))
    world.tick()

```

```
# Step 3 - release all controls
vehicle.apply_control(VehicleControl(throttle=0.0, brake=0.0,
                                     hand_brake=False))
for _ in range(5):
    world.tick()
```

### 11.3 Termination Conditions

1. Collision detected by CARLA collision sensor
2. Lateral deviation $|d_t| > 3.0\,\text{m}$ (lane violation)
3. Route complete: within 5 m of destination
4. Episode steps $\geq 2000$ (timeout)

## 12 Key Design Decisions Summary

1. **No privileged NPC positions.** All obstacle perception is from cameras and YOLO only. This is the core principle.

2. **HD map for ego state is realistic.** Every production AV stack uses pre-mapped HD maps for localisation. This is not cheating.

3. **True physical footprints on grid.** Obstacles are drawn at their estimated physical size. Free space literally means "my car fits here." No inflation heuristics needed.

4. **Ego at row 56, not row 63.** Asymmetric grid allocation: 56 pixels forward ($\approx$17.5 m), 8 pixels rear ($\approx$2.5 m). Forward awareness is more important.

5. **Stereo depth for front only.** Side and rear cameras use monocular ground-plane projection. High-precision depth is only critical forward.

6. **BC first, RL second.** Complex multi-objective reward landscape benefits from a warm-started policy. Framed as curriculum-guided imitation bootstrapping.

7. **Domain randomisation in Phase 3.** Weather, lighting, and NPC density randomised each episode to reduce sim-to-real gap.

8. **Wheel rotational inertia omitted.** $\approx$2.7% correction — smaller than CARLA physics noise. Noted as a known simplification.